

Exploring the Capability of ChatGPT for Cross-Linguistic Agricultural Document Classification: Investigation and Evaluation

Weiqiang Jin(✉), Biao Zhao, and Guizhong Liu

School of Information and Communications Engineering, Xi'an Jiaotong University, Innovation Harbour, Xi'an, Shaanxi, China
weiqiangjin@stu.xjtu.edu.cn, {biaozhao,liugz}@xjtu.edu.cn

Abstract. In the sustainable smart agriculture era, a vast amount of agricultural knowledge is available on the internet, making it necessary to explore effective document classification techniques for enhanced accessibility and efficiency. Over the past few years, fine-tuning strategies based on pre-trained language models (PLMs) have gained popularity as mainstream deep learning approaches, showcasing impressive performance. However, these approaches face several challenges, including a limited availability of training data, poor domain transferability, lack of model interpretability, and the challenges in deploying large models. Inspired by ChatGPT's significant success, we investigate its capability and utilization in the field of agricultural information processing. We explore various attempts to maximize ChatGPT's potential, including various prompting construction strategies, ChatGPT question-answering (Q&A) inference, and intermediate answer alignment technique. Our preliminary comparative study demonstrates that ChatGPT effectively addresses research challenges and bottlenecks, positioning it as an ideal solution for agricultural document classification. This findings encourage the development of a general-purpose agricultural document processing paradigm. Our preliminary study also indicates the trend towards achieving Artificial General Intelligence (AGI) for sustainable smart agriculture in the future. Code is available on Github (https://github.com/albert-jin/agricultural_textual_classification_ChatGPT).

Keywords: ChatGPT · Natural language processing · Agricultural document classification · Very large language model

1 Introduction

With the rapid development of the sustainable smart agriculture ecosystem, the quantity of various Internet news related to agricultural fields has seen an explosive increase. Artificial intelligence agricultural document classification enables the automatic management of this massive agricultural news and facilitates the

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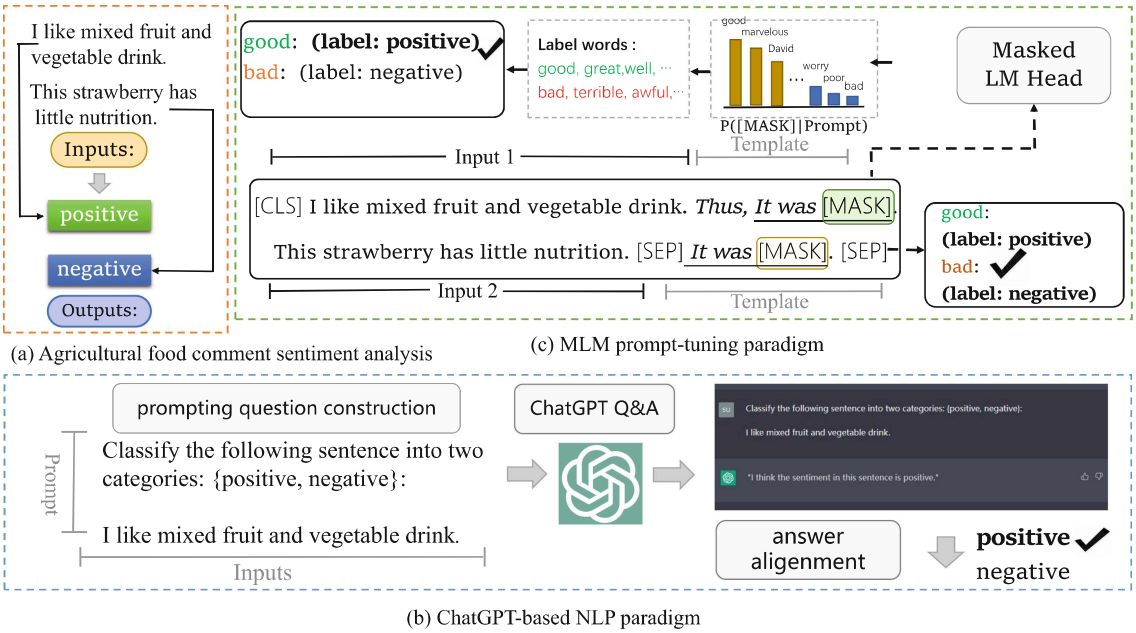


Fig. 1. Comparisons of ChatGPT-based NLP solutions with existing prompt learning paradigms by using an agricultural sentiment analysis example. Part (a) represents the task prototype of the agricultural sentiment analysis, while Part (b) and Part (c) illustrate the standard workflows of ChatGPT-based approaches and Masked LM prompt-tuning methods, respectively.

indexing of unstructured data, which is a crucial step towards agricultural digitization [22,25] and the agricultural Internet of Things development [9,26].

As a powerful very large-scale pre-trained language model (VLLM) for dialogue [19], ChatGPT has rapidly exhibited remarkable language comprehension abilities, attracting increasing attention in many cross-disciplinary researches [11,17,23]. Considering ChatGPT's impressive capabilities, it is natural to explore its potential in optimizing sustainable agricultural applications, including agricultural document classification.

In this paper, we investigate the potential of ChatGPT by focusing on the concise classification of agricultural text. This work introduces a novel NLP paradigm based on ChatGPT, which distinguishes it from existing methods. Figure 1 provides a clear illustration of the major similarities and differences between ChatGPT-based NLP paradigm (a) and MLM prompt-tuning paradigm (b). The prompt-tuning paradigm based on pre-trained language models (PLMs) can be divided into three primary procedures: template engineering, reasoning using pre-trained language models, and answer mapping engineering. Similarly, as depicted in part (b) of Fig. 1, natural understanding tasks based on ChatGPT can be organized into several phases [12,28]: 1) prompting question construction engineering; 2) ChatGPT question-answering (Q&A) inference; and 3) answer alignment (or normalization) engineering. To optimize the process, we considered several key factors: 1) designing task-specific inquiries to intuitively trigger

the understanding capability of ChatGPT; and 2) developing an accurate label mapping strategy from the outputs of ChatGPT to the final classified categories.

In our experiments, we initially obtained datasets from various agricultural sub-fields. These datasets were primarily sourced from Internet news articles that covered topics like insect pests, natural hazards, and agricultural market comments. By utilizing ChatGPT for tasks such as pest and disease identification, agricultural news categorization, and market comment analysis, we showcased how it can contribute to agricultural information management. Through a comprehensive set of experiments, including comparisons with state-of-the-art (SOTA) baselines, ablated experiments with advanced prompting strategies, ChatGPT triggered prompts and prompts from PromptPerfect[1], and several additional case studies, we thoroughly investigated and highlighted the distinct advantages of ChatGPT over other methods in advanced agricultural practices.

The primary contributions of this work can be summarized as follows:

- Motivated by the significant application success of ChatGPT, we conducted a preliminary study to explore the potentials of ChatGPT in agricultural document classification tasks and proposed a ChatGPT-based solution for agricultural document classification;
- ChatGPT-based document classification method outperforms existing PLM-based approaches when evaluated on multi-linguistic agricultural datasets, showcasing its impressive cross-lingual understanding capability;
- Our ChatGPT-based method that equipped with advanced prompts shows impressive performance gains, which demonstrates and highlights the importance and significance of the strategies about how to design prompts.
- The reliance of ChatGPT-based method subverts complex and power-intensive PLM-based methods, promising the low-cost and general artificial intelligence (AGI) techniques for smart agriculture. Code is released on Github[2].

2 Related Work

2.1 Agricultural Document Classification

Over the past decade, traditional machine learning approaches, such as SVM, CNNs, and RNNs, have dominated document classification research [24]. Azeez *et al.* utilized support vector machines (SVM) for regional agricultural land texture classification [1]. Dunnmon et al. demonstrated the superiority of CNNs over RNNs in sentiment classification of agriculturally-relevant tweets [6,13]. The introduced pre-trained language models like BERT [5] have significantly improved agricultural document processing tasks, gradually replacing traditional methods. Shi et al. used BERT to extract representative information from unlabeled sources, improving the efficiency of constructing corpora for agricultural

[1] PromptPerfect service: https://promptperfect.jinaai.cn/prompts.
[2] https://github.com/albert-jin/agricultural_textual_classification_ChatGPT.

news [18]. Jiang et al. classified French plant health bulletins using fine-tuned BERT, making the data easily searchable [9]. Cao *et al.* introduced an improved BERT-based sentiment analysis model for evaluating agricultural products using internet reviews [4]. The integration of large pretrained language models like BERT into agricultural document classification has shown promising results, advancing the field and improving efficiency and accuracy.

However, the current scale of existing PLMs is still not capable enough to meet the requirements for artificial general intelligence (AGI). There is an urgent need to accelerate the exploration of the capability of very large language models (VLLMs). Therefore, the tremendous interest in ChatGPT has sparked exploration into its potential and possibilities in the field of agriculture.

2.2 ChatGPT

ChatGPT [3,15], developed by OpenAI, is a leading conversational language model that provides expert knowledge in various fields. It is a disruptive revolution in research domains, extending beyond NLP and has made significant contributions to various application scenarios. As more and more users engage with ChatGPT, a multitude of novel applications emerge, showcasing its immense potential. ChatGPT has shown remarkable proficiency in multilingual translations, especially in high-resource languages such as various European and American languages [20]. ChatGPT's capabilities extend beyond translations to coding tasks, and it can assist in code debugging and generation. Recent empirical studies [2,7,25] have shown that ChatGPT can provide code snippets adhering to syntax and semantics in languages like Python, Java, and JavaScript. Additionally, ChatGPT has been explored for event extraction and information extraction tasks [7,8,21].

Overall, ChatGPT is a remarkable language model with a wide range of applications. It revolutionizes the accessibility and capabilities of large language models, opening up new possibilities for research and development. The groundbreaking nature of ChatGPT has sparked curiosity and exploration, including its potential applications in the agricultural field. While specific agricultural use cases are limited, researchers continue to investigate and uncover new possibilities for the sustainable agricultural development.

3 ChatGPT-Based Agricultural Document Classification

This paper focuses on the feasibility of applying ChatGPT to agricultural document classification. We present a series of exploratory experiments as a preliminary study on ChatGPT-based agricultural applications. As we know, there were no existing works that systematically utilized ChatGPT for document classification. To fill this gap, we discuss the general workflow for ChatGPT-based agricultural document classification. Based on extensive literature [21,28], we identify three phases that most ChatGPT-assisted applications can be divided into, as illustrated in Fig. 2:

i. Prompting Question Construction: This initial stage focuses on providing appropriate prompting strategies to input into ChatGPT;
ii. ChatGPT Q&A Inference: The second stage involves the ChatGPT question-answering reasoning procedure, which is transparent to us that treated as a black box;
iii. Answer Alignment: In the final stage, the natural language intermediate response is transformed into the target label within predefined categories.

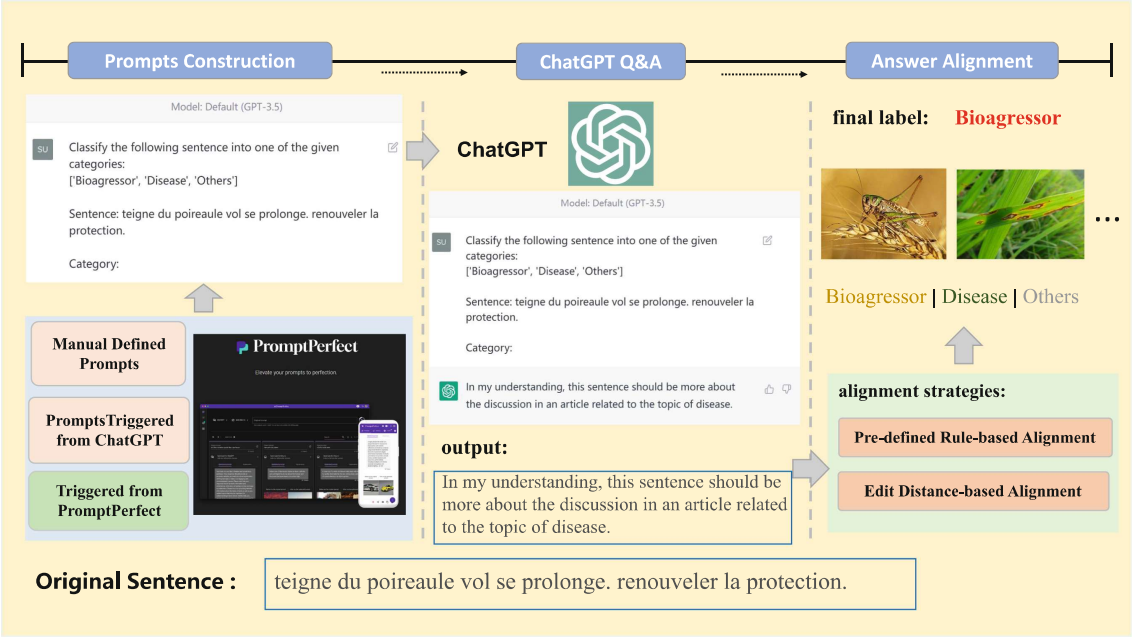


Fig. 2. The ChatGPT-based document classification framework is exemplified using the French Plant Health Bulletin dataset. It involves designing various prompts generation strategies (left), obtaining a response from ChatGPT (center), and aligning the intermediate answer to pre-defined categories (right).

Among these steps, while the question-answering inference conducted by ChatGPT remains static and beyond our modification, we can optimize the prompting construction and answer alignment engineering during our experiments. In the followings, we will introduce several novel strategies that we utilized to fully leverage the enormous potential and superiority of ChatGPT.

3.1 Prompt Question Construction

Prompts serve as a crucial component in determining the quality and relevance of generated outputs from language models. As inputs and guides for the model's output, the quality of prompts has a significant impact on the performance of the model. By leveraging prompt optimization engineering, users can unlock the full potential of large-scale language models and achieve better results with less effort and time. The engineering for constructing prompting question is widely

recognized as a complex task that involves extensive historical experience and manual trial-and-error [14,28]. As illustrated on the bottom-left of Fig. 2, we adopt several prompt generation strategies in our experiments, including: 1) manually defined prompts, 2) prompts triggered from ChatGPT, and 3) prompts triggered from the PromptPerfect platform.

Next, we discuss these prompt generation strategies in detail.

Table 1. The partial manually devised prompts. `[Res]` denotes the response provided by ChatGPT.

No.	prompting template
1	Classify the following sentence into one of the given categories: `[CATE]` \n Sentence: `[SENT]` \n Category: \t`[Res]`
2	Which categories do you think sentence: \n `[SENT]` \n belongs to, out of `[CATE]`? \n `[Res]`
3	.....

Manually Designed Prompts: Following the communication habits, we manually create several prompting templates. Table 1 showcases part of the designed templates. There are two essential elements that be provided to ChatGPT, denoting the sentence context and the pre-defined categories, respectively. For simplicity, we include two extra slots in the prompts to combine the respective mentions: slot `[SENT]` as the sentence, and slot `[CATE]` as the categories.

Note that not all manually designed templates are always the most suitable. It is necessary to select representative templates from the candidate set to be used for subsequent experiments. For this purpose, we utilize a data down-sampling evaluation as described in [2]. Specifically, we randomly select a predetermined number (100 by default) of samples from the specified dataset, Twitter Natural Hazards. Subsequently, we evaluate the performance of each template on this subset and choose the best-performed template as the most suitable manually defined prompt. Through a series of thorough down-sample comparison experiments, we selected the prompt for the subsequent experiments, which is depicted as follows.

```
> Categorize the given sentence into one of the provided
categories. Please provide a clear and concise response
that identifies the category of the sentence to allow for
categorizations.
The sentence is: [SENT].
The categories are specified in the [CATE].
The sentence to be classified is: {ChatGPT Response}.
```

Moreover, it should be noted that we have additionally included the following command in the prompts.

```
> ......, Please only answer the category.
```

This additional request is aimed at reducing unnecessary explanations generated by ChatGPT regarding its reply. Such explanations have the potential to disrupt the decision-making process for labeling subsequent text. We have also considered this factor when devising the following prompting methods.

ChatGPT Triggered Prompts: Taking cues from the extensive research in the field [10,27], we hypothesize that exploring ChatGPT directly might offer valuable insights into the creation of top-notch templates. With this in mind, we turn to ChatGPT itself, seeking its guidance on the recommended approaches for generating templates. Please note that the following human inquiries can serve as triggers for task-specific prompts:

```
> Provide five concise prompts or templates that can make
you deal with the [x] task.
```

, where the slot `[x]` represents task-specific types in the context.

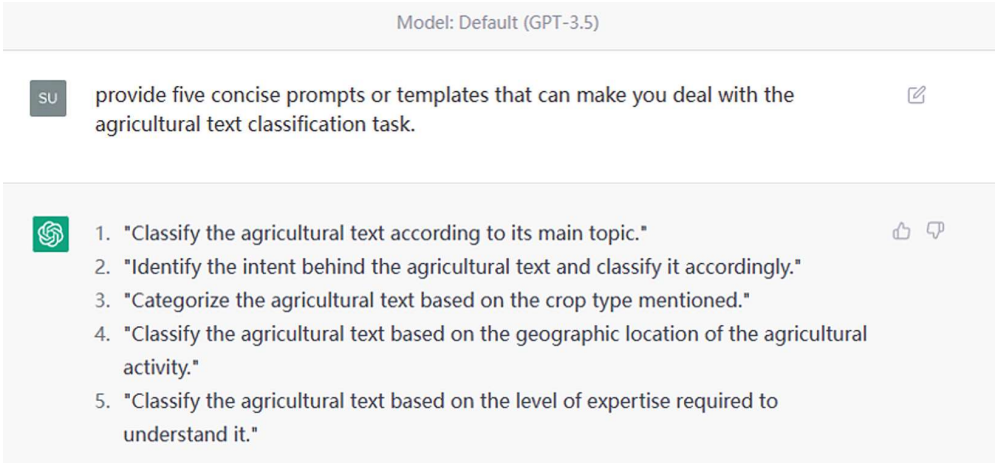


Fig. 3. Candidate prompt templates triggered by requests to ChatGPT (Model: GPT-3.5, Query Date: 2023.4.02).

As depicted in Fig. 3, we have intuitively formulated our request as follows:

```
> Provide five concise prompts or templates that can make
you deal with the agricultural document classification task.
```

The generated prompts demonstrate sensible and coherent semantic content, with discernible variations in their respective formats. Employing the sampling-based evaluation method described earlier, we have chosen the best-performing prompt to represent the ChatGPT-triggered prompts for subsequent comparative experiments. The selected prompt is as follows:

```
> Classify the agricultural text: [SENT] according to its
main topic [CATE].
```

Experiments prove that this like ChatGPT triggered prompting strategy can enhance the quality and accuracy of generated outputs, making them more useful and effective.

Prompts Triggered from PromptsPerfect: PromptPerfect[3] is a cutting-edge intelligent prompt optimizer for large-scale language models, which helps users craft prompts that yield exceptional results. PromptPerfect has the ability to fine-tune prompts and generate high-quality results for a wide range of applications. PromptPerfect can revolutionize users' experience with models like ChatGPT, MidJourney, and StableDiffusion.

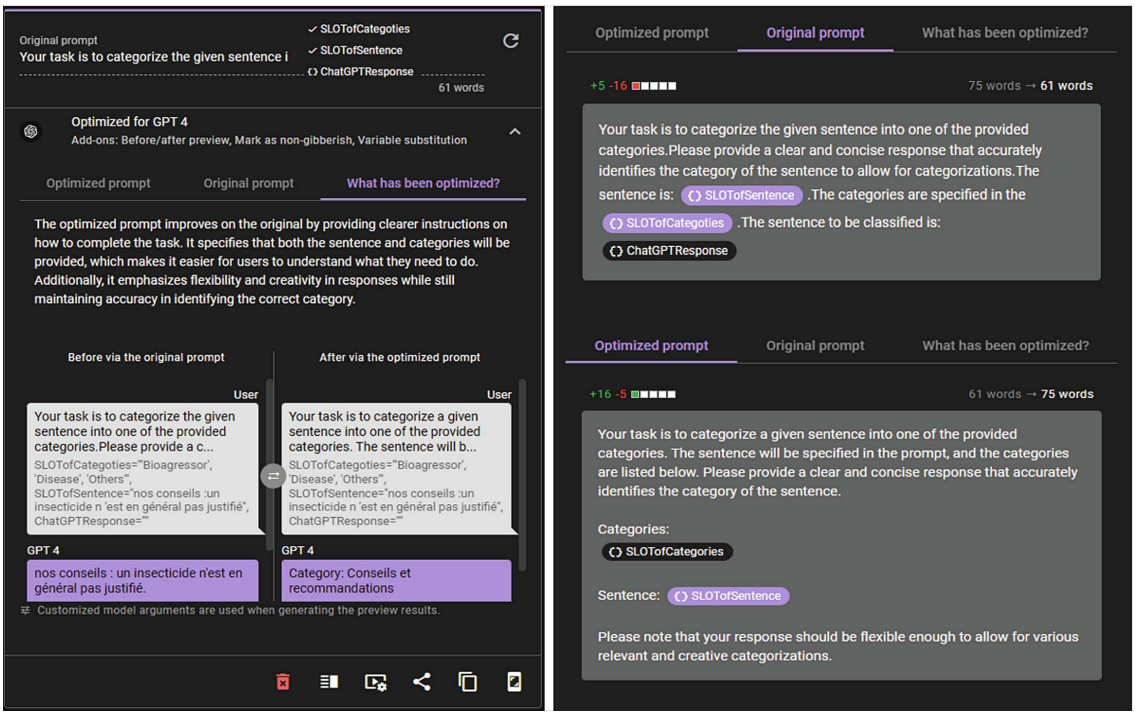


Fig. 4. The prompt optimization procedure of the PromptPerfect interface, exemplified by optimizing the manually designed prompt.

Here, we explore the impact of prompts optimized by PromptPerfect on the output quality of ChatGPT, taking advantage of PromptPerfect's advanced capabilities. Specifically, in the experiment, we simultaneously optimized the manually designed prompt and the ChatGPT-triggered prompt selected in the previous two steps. Figure 4 illustrates the optimization effects of the PromptsPerfect website service on the manually designed prompt.

The optimization result for the manually designed prompt is as follows:

[3] PromptPerfect service is available at https://promptperfect.jinaai.cn/prompts.

```
> Please classify the following agricultural document
according to its main topic:
Sentence: [CATE]
Categories: [SENT].
```

And the optimization result for the ChatGPT triggered prompt is as follows:

```
> Your task is to categorize a given sentence into one of
the provided categories. The sentence will be specified in
the prompt, and the categories are listed below. Please
provide a clear and concise response that accurately
identifies the category of the sentence.
Categories: [SENT]
Sentence: [CATE]
Please note that your response should be flexible enough to
allow for various relevant and creative categorizations.
```

Experimental results demonstrate that the overall performance of utilizing PromptPerfect for optimization are superior, surpassing the original prompts in a slight range of improvement.

3.2 ChatGPT Q&A Inference

The conversational prower of ChatGPT lies in its capacity to generate coherent sentence through sequence-to-sequence learning and the transformer architecture.

During inference, ChatGPT generates response by conditioning on a prompt and sampling words from a probability distribution. This distribution is computed using the softmax operation applied to the model's output. The output at each step is influenced by the previously generated tokens, enabling a generative process that produces coherent text.

Formally, the token generation process of ChatGPT can be expressed as follows:

$$p(y|x) = \prod_{i=1}^{L} p(y_i|y_1, ..., y_{i-1}, x) \tag{1}$$

where $\prod$ denotes the probability multiplication operator. The probability distribution $p(y_i|y_1, ..., y_{i-1}, x)$ represents the likelihood of generating token y_i at the t-th time step, given the preceding tokens $y_1, ..., y_{i-1}$ and the input prompt x. The length of the generated sequence is denoted as L.

To maintain consistency and enable independent thinking, we adopted a unique conversation thread for each prompt during the ChatGPT interaction process. This approach ensures that ChatGPT's responses are unaffected by the previous conversation history. By leveraging the user-provided information, ChatGPT consistently delivers optimal responses.

3.3 Answer Alignment

Different from traditional PLM-based classification models, the answers generated by ChatGPT do not directly correspond to predefined labels, posing challenges for subsequent output analysis and evaluation. Therefore, additional alignment strategies are required to convert these intermediate answers into final labels that can be used to compute various performance metrics. In our experiments, we designed and implemented two distinct alignment strategies: rule-based matching and similarity-based matching.

- **Rule-based matching:** This method utilizes predefined patterns or rules based on token attributes (e.g., part-of-speech tags) to match token sequences in unstructured textual data. During our experiments, we use the Matcher[4] in spaCy v3 to find the matched tokens in context.
- **Similarity-based matching:** This method computes the similarity between the generated answer and each label, selecting the label with the highest similarity score. First, we aggregate and synthesize commonly used expressions by ChatGPT for each category and establish a reference answer repository for each category. Then, we employ the Levenshtein distance algorithm to calculate the minimum edit distance between each reference answer and the input answer to be classified. The answer with the minimum distance is selected as the final category label.

To address the challenge of answer mapping, we adopt a pipeline approach that combines both rule-based and similarity-based matching strategies. We first use the rule-based strategy to parse the intermediate answer. If the category remains uncertain, we then employ the similarity-based strategy to calculate the similarity between the intermediate answer and each category's answer examples, selecting the most similar category as the final label.

4 Experiments

4.1 Setups

Datasets. We collected three datasets for agricultural document classification: **PestObserver-France**[5], **Natural-Hazards-Type**[6], and **Agri-News-Chinese**[7], ranging from different types of categories (e.g. plant diseases, insect pests, and twitter natural hazards) and numbers of categories to various languages, including French, English, and Chinese. Note that Natural-Hazards-Type is a disaster categories classification dataset re-organized from the original sentiment classification dataset, Natural-Hazards-Twitter[8]. The details are depicted in Table 2.

[4] Spacy can be accessed on https://spacy.io.
[5] PestObserver-France download: https://github.com/sufianj/fast-camembert.
[6] https://github.com/albert-jin/agricultural-_textual_classification_ChatGPT.
[7] More details can be accessed from http://zjzx.cnki.net/.
[8] https://github.com/Dong-UTIL/Natural-Hazards-Twitter-Dataset.

Table 2. The statistical meta-information of these datasets. N-train, n-test, n-label denote the count of training samples, test samples, and categories, respectively.

Datasets	n-train	n-test	language	labels	n-label
PestObserver-France	322	80	French	Bioagressor, Disease, Others	3
Natural-Hazards-Type	5000	1000	English	Hurricane, Wildfires, Blizzard, Floods, Tornado	5
Agri-News-Chinese	52000	6500	Chinese	农业经济, 农业工程, 水产渔业, 养殖技术, 林业, 园艺, 农作物	7

Baselines. To conduct a comprehensive comparison with previous machine learning methods, such as feature engineering-based method, representation learning method, PLM-based fine-tuning and prompt-tuning. We selected a series of representative approaches as baselines. These baselines include SVM [1], TextRNN [13], BERT-based fine-tuning [5] and T5-based prompt-tuning methods [16], each representing a different learning paradigm mentioned above. For TextRNN, we utilized the pre-trained word vectors from GloVe[9] as embeddings. The PLM models used in our study are the "bert-base-uncased" and "t5-base" versions obtained from facebook/huggingface.io[10].

Metrics. In agricultural document classification tasks with multiple labels, accuracy and F1-score are commonly used metrics. Accuracy measures the proportion of correctly predicted samples among all predicted samples, while F1-score considers precision and recall.

Experimental Environment. The meta experimental settings can be summarized as follows: The experimental hardware environment consists of an Intel Core i9-9900k CPU and a single Nvidia GPU, specifically the GTX 1080Ti. The code implementation is done in Python 3.7 and PyTorch 1.9.0.

4.2 Main Results

Table. 3 details the comparison results of our method and these state-of-the-art baselines on the three cross-linguistic datasets. For comparison simplicity, we regarded the approach which adopts the manually defined prompts as the vanilla ChatGPT-based classification method and took it for comparison.

Through the comparative experimental results, our approach exhibits competitive performance across all datasets. ChatAgri demonstrates a significant performance advantage of 10% to 20% compared to traditional SVM and TextRNN methods. Furthermore, our approach shows impressive performance compared to the latest methods based on large-scale pre-trained models. Compared to PLM BERT fine-tuning and PLM T5 prompt-tuning methods, our approach improves accuracy or weighted F1 scores. For example, compared to BERT fine-tuning, our approach achieves an accuracy improvement of approximately

[9] GloVe embedding: https://nlp.stanford.edu/projects/glove/.
[10] Huggingface transformers: https://huggingface.co/docs/transformers/index.

Table 3. Performance statistics of the baselines and ChatAgri on the adopted datasets. The best and second-best scores across all models are respectively highlighted in bold and underlined. [ChatGPT Query Date: March 16, 2023].

Learning Paradigms	Baselines	PestObserver -France		Natural-Hazards -Type-English		Agri-News -Chinese	
		Acc	Weighted -F1	Acc	Weighted -F1	Acc	Weighted -F1
Feature Engineering	SVM	0.672	0.645	0.811	0.811	0.623	0.622
Word Representation	TextRNN	0.707	0.697	0.931	0.931	0.812	0.801
PLM-based Fine-tuning	BERT-based fine-tuning	0.736	0.714	0.945	0.945	0.826	0.819
PLM-based Prompt-tuning	T5-based prompt-tuning	<u>0.764</u>	<u>0.753</u>	<u>0.966</u>	<u>0.966</u>	<u>0.859</u>	<u>0.854</u>
ChatGPT-based Prompt QA	(Our Method)	**0.794**	**0.789**	**0.978**	**0.978**	**0.863**	**0.856**

5.8% and 3.7% on the PestObserver-France and Agri-News-Chinese datasets, respectively. Compared to the T5 prompt-tuning method, our approach achieves slightly higher accuracy on the Agri-News-Chinese and Natural-Hazards-Type datasets, approximately 3.0% and 1.1% respectively.

In addition, ChatGPT performs excellently in cross-language understanding. Unlike traditional PLM-based methods, ChatGPT utilizes comprehensive and high-quality training corpora covering most languages used in various countries. Moreover, the enormous parameter size of ChatGPT enables it to retain and master more language knowledge, not limited to English. Thus, ChatGPT exhibits significant superiority over traditional PLM models (such as BERT, RoBERTa, and BART) in terms of cross-language understanding.

4.3 Improved with ChatGPT Triggered Prompts

Here, we discuss the performance influence of ChatGPT triggered prompting strategy. The overall results of ChatGPT equipped with ChatGPT triggered prompts are presented in Table 4. Specifically, from the third and fourth columns of Table 4, our method that equipped with ChatGPT triggered prompts yielded an approximately 2.1% accuracy and 1.4% weighted-F1 improvements on PestOb-server-France over that using manually constructed prompts, and outperforms the basic counterpart by a performance margin about 1.2% both of accuracy and weighted-F1 on Agri-News-Chinese.

However, there was no significant improvement in performance for the Natural-Hazards-Type. We speculate that this may be attributed to the characteristics of the dataset, where the semantics related to classifying natural disaster categories are prominently evident, often relying solely on a few fixed

trigger words and phrases. Consider the following sentence found in the dataset: "Matthew charges toward Jamaica, Haiti, Cuba: Islanders stock up on food, water and other supplies." In this sentence, the referenced word "Matthew" is intrinsically associated with the document topic of an American hurricane disaster.

Table 4. The performance comparison of our method using the manually constructed prompts and the prompts generated by ChatGPT. The colored numbers shown in percentage indicated the increased performances.

Prompting Strategies		Manually Defined - Prompts (Vanilla)	ChatGPT Triggered - Prompts
PestObserver -France	Accuracy	0.794	**0.815** ↑ **2.1%**
	Weighted-F1	0.789	**0.812** ↑ **1.4%**
Natural-Hazards-Type	Accuracy	**0.978**	**0.978** ↑ **0.0%**
	Weighted-F1	**0.978**	**0.978** ↑ **0.0%**
Agri-News -Chinese	Accuracy	0.863	**0.874** ↑ **1.1%**
	Weighted-F1	0.856	**0.867** ↑ **1.1%**

One additional notable experimental phenomenon is that prompts obtained through ChatGPT tend to be concise and succinct, while manually crafted prompts often become excessively lengthy due to the inclusion of a series of constraining conditions. This finding further substantiates the lack of a direct relationship, in theory, between the length of the provided prompt and the quality of feedback and classification performance achieved by ChatGPT. Accordingly, appropriately providing brief prompts can still effectively stimulate ChatGPT's exceptional language comprehension capabilities.

In summary, with the help of the advanced prompts triggered by ChatGPT, the proposed classification method achieves impressive performance improvements and shows its significance and effectiveness. Furthermore, designing appropriate templates to stimulate the potential generative capacity of ChatGPT is an valuable research area that warrants thorough exploration and discussion.

4.4 Improved with PromptPerfect Prompt Optimization

Here, we provided a detailed analysis towards the PromptPerfect's prompt optimization effect on our ChatGPT-based classification method. Table 5 statistics the comparative experimental results between the use of PromptPerfect to optimize prompts before and after.

In this table, the first and second groups show the increased performance of manually constructed prompts and ChatGPT triggered prompts equipped with

PromptPerfect, respectively. It can be clearly observed that prompts optimized through PromptPerfect facilitate ChatGPT in providing more accurate feedback responses and achieving higher metrics. Although there was only a slight improvement in the Natural-Hazards-Type dataset, it can be mainly attributed to the fact that the dataset already had high prediction performance, approaching saturation. For instance, after being optimized with PromptPerfect, our methods, based on the two prompting designs, showed an averaged improvements of over 1% on the PestObserver-France dataset. Furthermore, our method utilizing manually constructed prompts that enhanced by PromptPerfect exhibited an increase of 2.6% in accuracy and a increase of 2.3% in weighted-F1 on the Agri-News-Chinese dataset.

Table 5. The performance statistics of our method enhanced by the PromptPerfect tool. The first group and the second group respectively represent the two designed prompts, manually defined prompts and ChatGPT triggered prompts.

Prompting Strategies	PestObserver -France		Natural-Hazards -Type		Agri-News -Chinese	
	Acc	Weighted -F1	Acc	Weighted -F1	Acc	Weighted -F1
Manually Defined - Prompts (Vanilla)	0.794	0.789	0.978	0.978	0.863	0.856
Manually Defined - Prompts + PromptPerfect	0.808 ↑1.4%	0.802 ↑1.3%	0.982 ↑0.4%	0.982 ↑0.4%	0.889 ↑2.6%	0.879 ↑2.3%
ChatGPT Triggered - Prompts	0.815	0.812	0.978	0.978	0.874	0.867
ChatGPT Triggered - Prompts + PromptPerfect	0.832 ↑1.7%	0.827 ↑1.5%	0.980 ↑0.2%	0.980 ↑0.2%	0.886 ↑1.2%	0.876 ↑0.9%

The experiments conducted have provided compelling evidence of the significant and profound influence of different designed prompts on the performance of agricultural document classification. This finding can be extrapolated to various other language understanding tasks as well. We believe that meticulously crafted prompts possess the capability to guide AI models in generating outputs of higher quality and greater value, thereby facilitating more intelligent information processing. Recently, Baidu's founder, Robin Li, stated that in ten years, prompt engineering would encompass half of the world's jobs, highlighting the immense employment opportunities in this field[11]. As we venture into the future of artificial intelligence, the proficiency in prompt design will determine the boundaries

[11] Baidu Ernie Bot: https://wenxin.baidu.com/wenxin/nlp.

of advancement for large-scale language models. Good prompts have the potential to effectively unlock greater potential in large-scale language models like the ChatGPT series, thus enabling their wider application and impact in real-world scenarios.

5 Prominent Cross-Linguistic Supports

To evaluate the level of support that ChatGPT and traditional PLMs provide for different languages, especially low-resource languages, we selected three typical French document fragments from the PestObserver-France dataset. We tokenized these fragments using the tokenizer provided by the official transformers library[12]. The intermediate tokenization results are shown in Table 5.

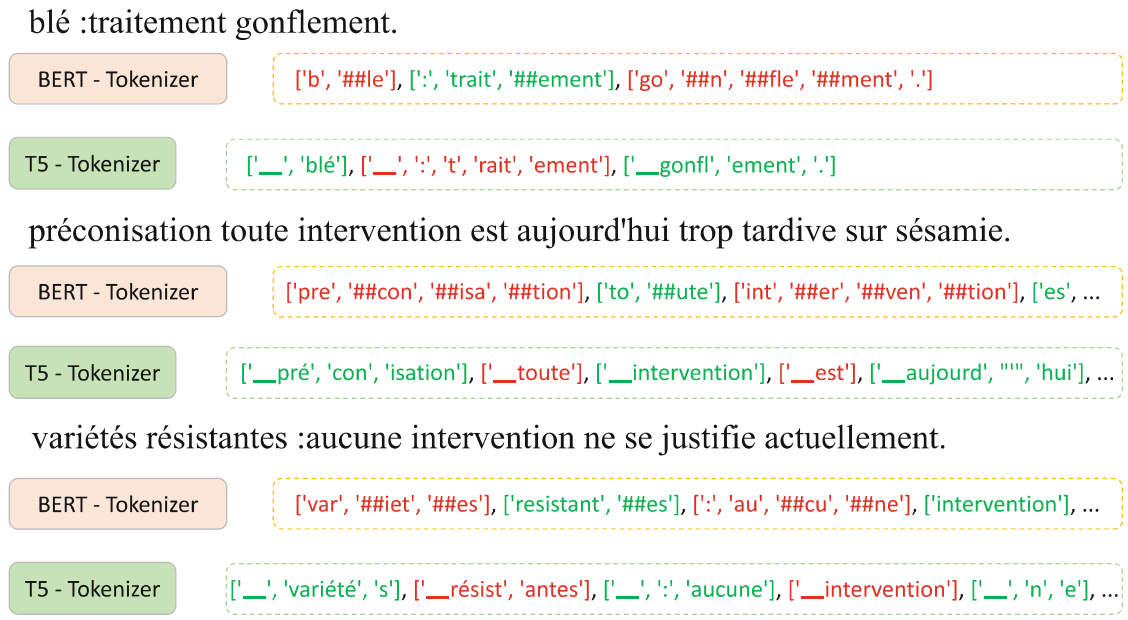


Fig. 5. Three tokenized cases generated by the tokenizers of BERT and T5 on the PestObserver-France dataset. Each group of the same color represents a independent word segmentation result.

From the table, it is evident that BERT and T5 have insufficient vocabulary coverage and support for small languages like French. Specifically, due to the limited vocabulary size during the pre-training phase, most of the French words in these examples are incorrectly segmented. For instance, in the first example, the French word `gonflement` is incorrectly segmented into four subword units with semantic ambiguities: `'go'`, `'##n'`, `'##fle'`, and `'##ment'`. In the second example, `intervention` is segmented into four unrelated individual words: `'int'`, `'##er'`, `'##ven'`, and `'##tion'`. Similarly, in the final example, `variétés` is segmented into `'var'`, `'##iet'`, and `'##es'`. Such inappropriate

[12] https://huggingface.co/docs/transformers/v4.29.1/en/index.

segmentations greatly affect the performance of subsequent language models in semantic understanding during fine-tuning. The primary reason is that the current mainstream pre-trained language models are still limited by the training data, which prioritizes scale and thus restricts their effective and comprehensive coverage of various small languages.

In contrast, one significant advantage of ChatGPT lies in its enormous parameter size. ChatGPT is based on a corpus of 800 billion words (or 45 terabytes of textual data) and contains 175 billion parameters. The 800 billion refers to the training data used for ChatGPT, while the 175 billion model parameters can be considered as the knowledge and representation distilled from this training data. Previously, the largest models only had parameters in the billions, not in the hundreds of billions. Therefore, this larger parameter size enables ChatGPT to capture more complex language patterns and relationships, thereby improving the accuracy of complex natural language processing tasks.

6 Conclusion

Agricultural document classification is a crucial step in managing the vast and ever-increasing amount of agricultural information for sustainable agricultural application. However, mainstream PLM-based classification approaches face several challenges, such as dependency on well-annotated corpora, cross-linguistic transferability. The emergence of ChatGPT, a leading OpenAI conversational technology, has brought a turning point to this dilemma. In this study, we propose a novel ChatGPT-based agricultural document classification framework, to explore the feasibility and potential of using ChatGPT for agricultural document classification. We conducted extensive experiments on datasets that included various languages, comparing our ChatGPT-based classification approach with existing baselines from different learning paradigms. We also developed several effective prompting strategies to further stimulate the generative capability of ChatGPT. The experimental results demonstrate the superiority of ChatGPT in agricultural document classification. Our empirical exploration has opened up new milestones for the development of techniques of ChatGPT-based agricultural information management, without exaggeration. We look forward to proposing more ChatGPT-based applications in sustainable agricultural development, which will help promote future agricultural digital transformation and sustainable agricultural development simultaneously.

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